

Research Proposal: A Power Side Channel Exploiting AVX Instructions

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1 Background

Side-channel attacks are a class of security exploits where an attacker gains sensitive information not from direct intrusion into a system, but rather from the information leaked by the physical implementation of a system. These attacks exploit the physical characteristics associated with the computation, such as time taken to execute certain operations, power consumption, electromagnetic leaks, or even sound to extract sensitive information.

Power side-channel attacks [1] specifically exploit variations in power consumption during the execution of a program. As different instructions and operations consume varying amounts of power, an attacker can analyze these power consumption patterns to infer sensitive information about the data being processed. This form of attack has been proven effective against a wide range of systems, from embedded devices to high-performance servers.

Advanced Vector Extensions (AVX) are extensions to the x86 instruction set architecture for microprocessors from Intel and AMD. Introduced to enhance performance in floating-point-intensive applications, AVX instructions allow for operations on large vectors of data in a single instruction, significantly increasing computational efficiency. Many popular linear algebra libraries today leverage the AVX instruction set to achieve substantial speed-ups in calculations. Notably, Eigen, one of the most prominent linear algebra libraries [2], and TensorFlow, a renowned open-source machine learning library developed by Google Brain, designed for deep learning in Python [3], employ AVX instructions to optimize calculation times. However, the use of AVX instructions also leads to increased power consumption, which can potentially be exploited by power side-channel attacks.

The study of power side-channel attacks on systems using AVX instructions is a relatively new field, but one with significant implications for the

security of a wide range of systems. As more systems adopt AVX instructions for their computational efficiency, understanding and mitigating the potential security risks associated with their power consumption becomes increasingly important.

2 Related Work

Building upon the line of research in power side-channel attacks, Liu et al. [4] introduced a novel type of side-channel attack known as the frequency throttling side-channel attack. This attack exploits the workload-dependent CPU frequency adjustments performed by the power management architecture of modern processors. The discovery of this attack further underscores the need for comprehensive security measures that take into account the physical characteristics of hardware implementations.

In another significant work, Wang et al. [5] proposed a new approach to power side-channel attacks. They demonstrated how these attacks can be turned into remote timing attacks on x86 systems, opening up new avenues for research in the field of side-channel attacks.

Further delving into the realm of software-based power side-channel attacks on x86, Lipp et al. [6] presented an enhanced technique to distinguish between instructions, operands, and data. Their research, titled "PLATYPUS", showcases that the Running average power limit (RAPL) interface offers a resolution that is sufficient for executing practical attacks on Intel CPUs. This work provides valuable insights into the intricacies and potential vulnerabilities associated with software-based power side-channel attacks.

3 Goals

Our primary goal is to design a side-channel attack that capitalizes on the power consumption patterns associated with AVX operations. By doing so, we aim to uncover potential leaks of sensitive information, particularly concerning matrices. A significant aspect of this endeavor is to extract as much information as possible about a matrix, understanding its structure and properties. The motivation behind this focus on matrices stems from their critical role in a vast array of computational tasks. Matrices are fundamental to a multitude of mathematical and physics-based calculations. They serve as the foundation for numerous algorithms and are indispensable in fields like machine learning, especially in deep learning architectures such as convolutional neural networks. Beyond theoretical applications, matrices play a

pivotal role in real-world scenarios. For instance, they are crucial in physical simulations in aeronautics, seismic tests for earthquake predictions and optimization problems in logistics. Given the ubiquity of matrices in both theoretical and practical applications, identifying vulnerabilities associated with them could have significant consequences. Additionally, a secondary, but equally crucial objective, is to delineate the precise relationship between AVX instructions and their power consumption. By achieving a deeper understanding of this relationship, we hope to provide insights that can guide future security measures and optimizations.

4 Preliminary Results

The preliminary results shed light on the relationship between the power consumption of AVX multiplication operations and the characteristics of their inputs. As illustrated in Figure 1, a discernible correlation emerges between the Hamming weight of the input to the multiplication instruction and the resultant power consumption. Specifically, as the Hamming weight increases, there is a rise in power consumption.

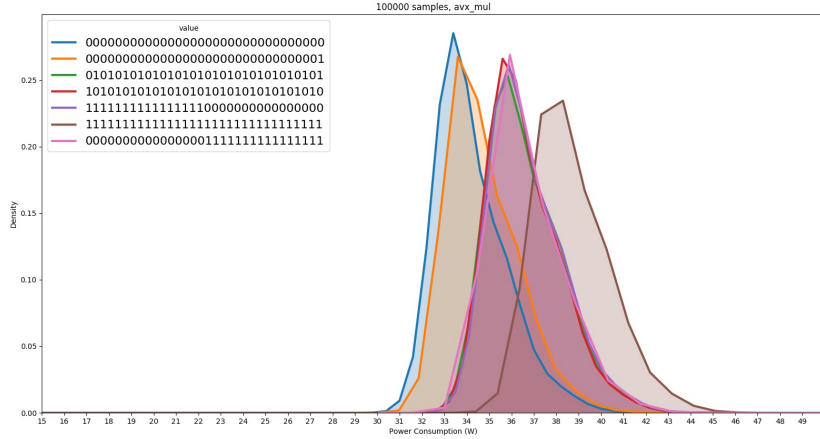


Figure 1: Power consumption of AVX multiplication with different inputs. The input to the AVX multiplication comprised two 256-bit vectors, each consisting of eight 32-bit segments, as represented by distinct colors.

Further investigations were conducted to understand the power consumption behavior in relation to matrix sparsity across two widely adopted libraries: Eigen and TensorFlow. A clear trend was observed in both cases: matrices with higher sparsity levels consumed less power during multiplication operations. Given TensorFlow’s reliance on the Eigen library for certain

operations, the power consumption patterns observed in TensorFlow mirrored those of the Eigen experiment, reinforcing the potential vulnerabilities associated with matrix sparsity in these libraries.

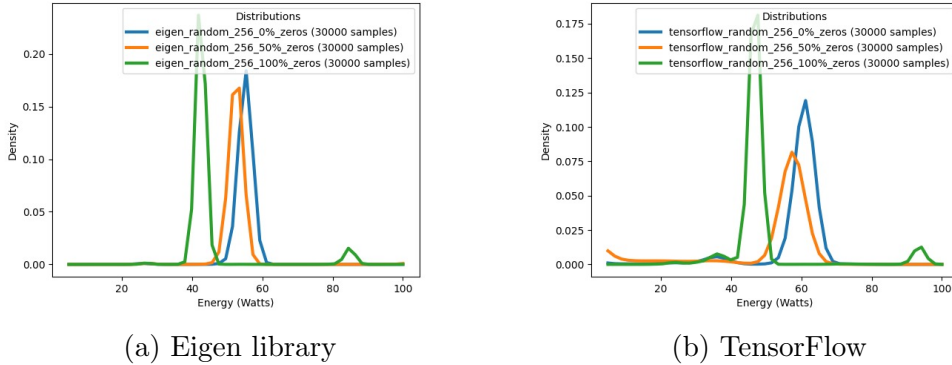


Figure 2: Comparative power consumption during matrix multiplication for different libraries highlighting the influence of matrix sparsity.

References

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